

Challenge Topics: Receiver

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Notes on the Objectives of the Activity

Objectives of the Activity and Brief Explanation of the Components Used

1) Photodiode Response and Reverse Bias Operation

Notes on the Objectives of the Activity

1.1) Photodiode Operation

The photodiode is a sensor that transforms the infrared light into an electrical current. When the IR beam hits the diode, the light energy creates electron-hole pairs, which generate a small current. This current is proportional to how much light the photodiode receives.

We use the photodiode in reverse bias because this makes the response faster and increases sensitivity. With reverse bias, the depletion region becomes larger, the capacitance decreases, and the diode reacts quickly to the pulses sent by the transmitter.

When the beam is present, the photodiode produces a pulsed signal.
When the beam is blocked, the current drops to almost zero (dark current).

1.2) Reverse Bias Configuration

The photodiode is typically operated under reverse bias (positive terminal connected to the N-side/cathode; negative to the P-side/anode) to optimize its performance as a light detector:

Increased Speed: Reverse bias widens the depletion region. A wider depletion region reduces the junction's capacitance. Lower capacitance allows the diode to respond more rapidly to changes in light, which is crucial for detecting the high-frequency pulses from the transmitter.

Enhanced Sensitivity: The intrinsic current flow without light (known as dark current) is minimized. This ensures that the increase in current generated by the IR beam is more pronounced, significantly improving the signal-to-noise ratio and the overall sensitivity of the receiver.

1.3) Output Signal Behavior

The IR barrier system operates using a pulsed signal.

Beam Present: The diode outputs a pulsed current that matches the exact frequency of the transmitter.

Beam Blocked: The IR light ceases to strike the photodiode. The photocurrent drops abruptly to the low level of the dark current.

2) Transistor Amplifier Stage

After the photodiode, the signal is still very weak, so two NPN transistors are used to amplify and clean the signal.

Using two stages helps to:

Increase the total gain remove noise shape the pulses so the next circuit (the 567) can read them properly

The second transistor works slightly in saturation to guarantee clean digital levels, improving noise immunity.

3) NE/LM567 – PLL Function

The 567 is the frequency decoder of the receiver.
Its job is to check if the correct IR signal is present.

Pin 8 LOW → The PLL is locked. Beam intact.

Pin 8 HIGH → The PLL is unlocked. Beam blocked → alarm must activate.

The decoding frequency calculated for the 567 is:

$$f \approx 4.13 \text{ kHz}$$

4) Alarm Oscillator (NE555)

The 555 for the alarm is controlled by Pin 8 of the 567.

When the beam is intact, Pin 8 is LOW → the 555 stays OFF.

When the beam is blocked, Pin 8 goes HIGH → this enables Pin 4 of the 555 → alarm turns ON.

The 555 generates an audible tone around 152.4 Hz with duty cycle $\approx 50\%$.

5) Circuit Optimization – 400 Hz

To obtain a frequency close to **400 Hz** on the 555 alarm oscillator, the values were adjusted to:

$$R_A = 100 \, \Omega$$

$$R_B = 18 \, \text{k}\Omega$$

$$C = 100 \, \text{nF}$$

This modification brings the output frequency close to the required 400 Hz.

Calculations and answers:

$$f_{alarm} = \frac{144}{((1 \times 10^3) + 2 \times (100 \times 10^3)) \times (47 \times 10^{-9})}$$

$$f = \frac{144}{(1000 + 200000) \times (47 \times 10^{-9})}$$

$$f = \frac{144}{(201000) \times (47 \times 10^{-9})}$$

$$f = \frac{144}{9.447 \times 10^{-3}}$$

$$f_{alarm} \approx 152.4 \text{ Hz}$$

Resistor RA: 1K Ω
Resistor RB: 100K Ω
Capacitor C: 47 nF

$$D = \frac{R_A + R_B}{R_A + 2R_B} \times 100$$

$$D = \frac{1K + 100K}{1K + 2 \times 100K} \times 100$$

$$D = \frac{101K}{201K} \times 100$$

$$D \approx 0.50248 \times 100$$

$$D \approx 50.25$$

$$\text{frequency} \approx 4.13 \text{ kHz}$$

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$$R_1 = 10K$$

$$C_1 = 22n$$

b) Calculate 567 Tone decoding Frequency (Refer to datasheet)

$$f_0 = \frac{1}{0.1 \times R_1 \times C_1} \quad \left\{ \begin{array}{l} f_0 = \frac{1}{1.1 \times (10 \times 10^3) \times (22 \times 10^{-9})} \\ f_0 = \frac{1}{1.1 \times 220 \times 10^{-6}} \end{array} \right.$$

R:

$$f_0 = \frac{1}{242 \times 10^{-6}}$$

$$f_0 \approx 4132.23 \text{ Hz}$$

$$f_0 \approx 4.13 \text{ kHz}$$



